

Genome Exploration II

Basic Genome Structure and Sequence Exploration Using Galaxy

Name	Suan, Jade Angela B.
Course and Section	BIO 300 - B
Date	August 12, 2026
Instructor	Mr. Abner Bucol
Species name	<i>Hipposideros armiger</i> (Great roundleaf bat)

Part 1 — Reconfirm the Genome Identity

Species name	<i>Hipposideros armiger</i> (Great roundleaf bat)
NCBI Assembly accession	GCF_001890085.2
Assembly level (if known)	Scaffold
FASTA filename in Galaxy	Hipposideros_armiger_genome_original.fna.gz
Genome source/database	NCBI RefSeq
Approximate file size	637.6 MB

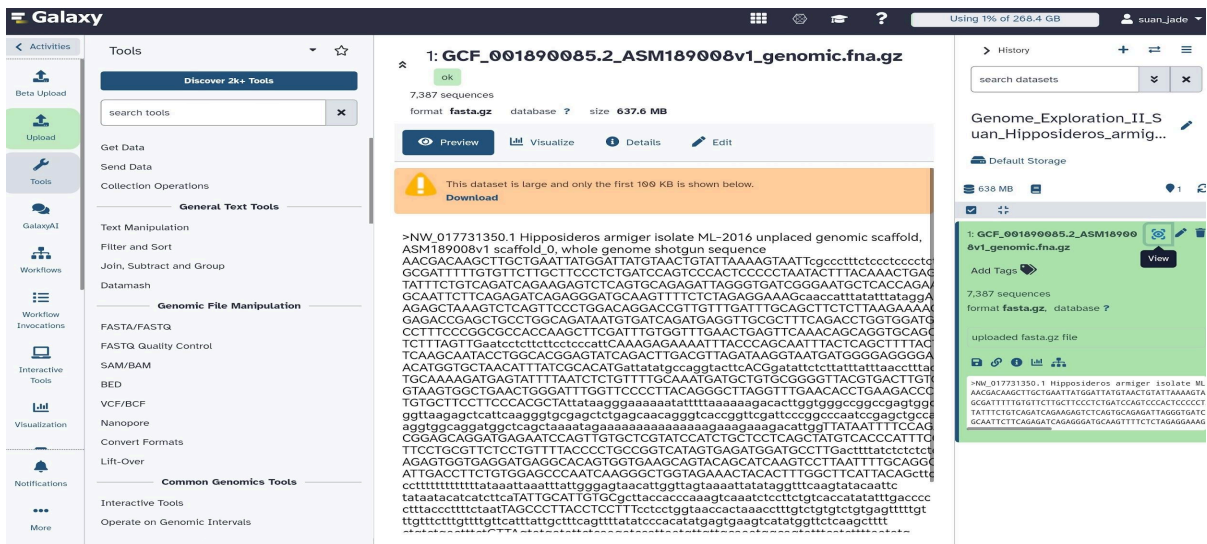


Figure 1. Uploaded *Hipposideros armiger* genome assembly in Galaxy

Part 2 — Generate Basic Assembly Statistics

Metric	Values	Interpretation
Total assembly length (bp or Mb)	2,236,326,314	Standard genome size for a bat
Number of sequences/contigs/scaffold	7,387	Few scaffolds - good overall contiguity
Minimum sequence length	877	Shortest sequence; small fragments still present
Maximum sequence length	1,2387,654	Long scaffold- shows excellent regional assembly
Mean or average sequence length	302,738	Long average - fewer short fragments overall
N50	2,328,177	Half the genome in scaffolds ≥ 2.33 Mb - very strong contiguity
L50	272	Only 272 scaffolds = 50% of total - well connected
GC content (%)	35.93	AT-rich - typical for mammalian genomes

Galaxy

Using 5% of 268.4 GB

suan_jade

Activities

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FASTQ Quality Control

SAM/BAM

BED

VCF/BCF

Nanopore

Convert Formats

Lift-Over

Common Genomics Tools

Interactive Tools

2: Hipposideros_armiger_Assembly_Statistics_gfastats

30 lines 2 columns

format tabular database ? size 680 b

Preview Visualize Details Edit

Column 1	Column 2
Scaffold L50	272
Scaffold N50	2328177
Scaffold L90	1069
Scaffold N90	490667
Scaffold len_max	12387654
Scaffold len_min	877
Scaffold len_mean	392738
Scaffold len_median	3500
Scaffold len_std	986243
Scaffold num_A	575630882
Scaffold num_T	575366364
Scaffold num_C	401813281
Scaffold num_G	401727540
Scaffold num_N	281788247
Scaffold num_bp	2236326314
Scaffold num_bp_not_N	1954538067
Scaffold num_seq	7387

History

search datasets

Genome_Exploration_II_Suan...

Default Storage

12 GB 16 2 2

Scaffold L90 1069

Scaffold N90 490667

Scaffold len_max 12387654

1: Hipposideros_armiger_genome_original.fn.gz

Add Tags

7,387 sequences

format fasta.gz database ?

uploaded fasta.gz file

>M_017731350.1 Hipposideros armiger_genome_original.fn.gz

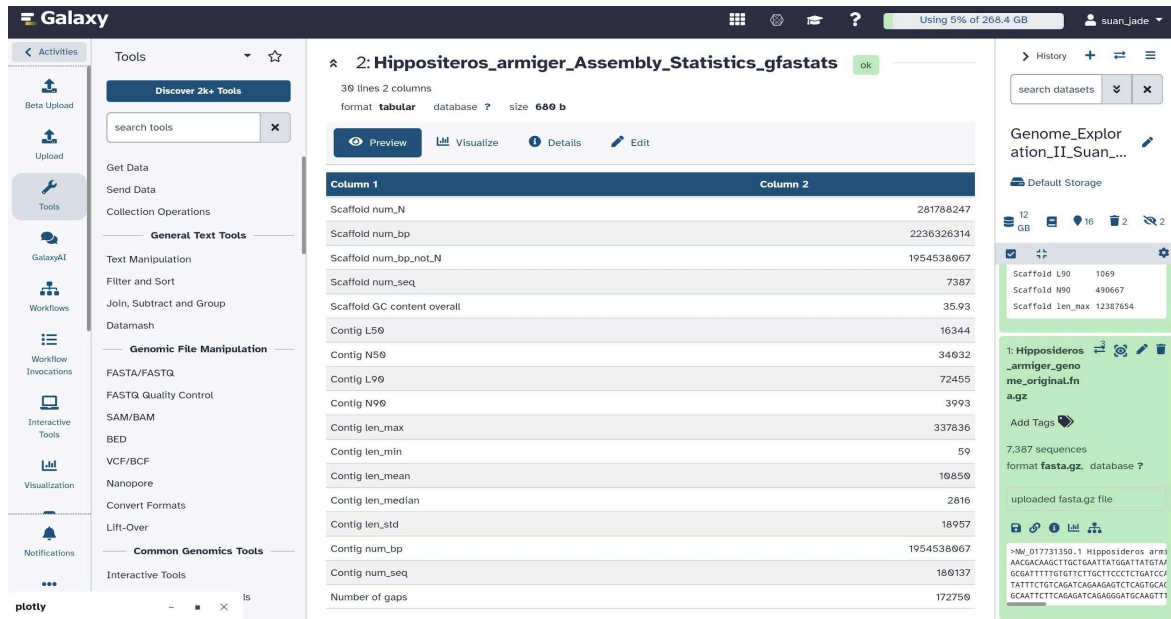


Figure 2. Fasta Statistics output showing assembly metrics for the *Hipposideros armiger* genome

Part 3 — Examine Sequence-Length Structure

Rank	Sequence ID	Length (bp)
1	NW_017731350.1	12,387,654
2	NW_017731353.1	9,904,920
3	NW_017731352.1	9,617,682
4	NW_017731354.1	9,428,270
5	NW_017731358.1	9,427,274

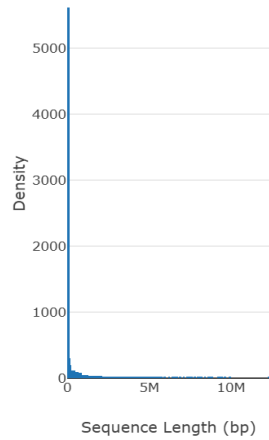


Figure 3. Histogram of showing the distribution of scaffold lengths in the *Hipposideros armiger* genome assembly



Figure 4. Sorted scaffold-length table showing the 5 longest scaffolds of the *Hipposideros armiger* genome ranked from largest to smallest

The genome is made up of very long sequences and many shorter ones. The top five scaffolds are all extremely large over 9.4 Mb with the longest at 12.39 Mb. These huge sequences show that big parts of the genome were assembled very well. After the top five, the rest of the sequences get shorter but most are still long at around 300,000 bp on average.

Part 4 — Perform a Simple Length-Filtering Experiment

Metric	Original genome	After ≥10kb filter
Total length	2,236,326,314	2,236,190,579
Number of sequences	7,387	7,247
Maximum length	12,387,654	12,387,654
N50	2,328,177	2,328,177
GC content	35.93%	35.93%

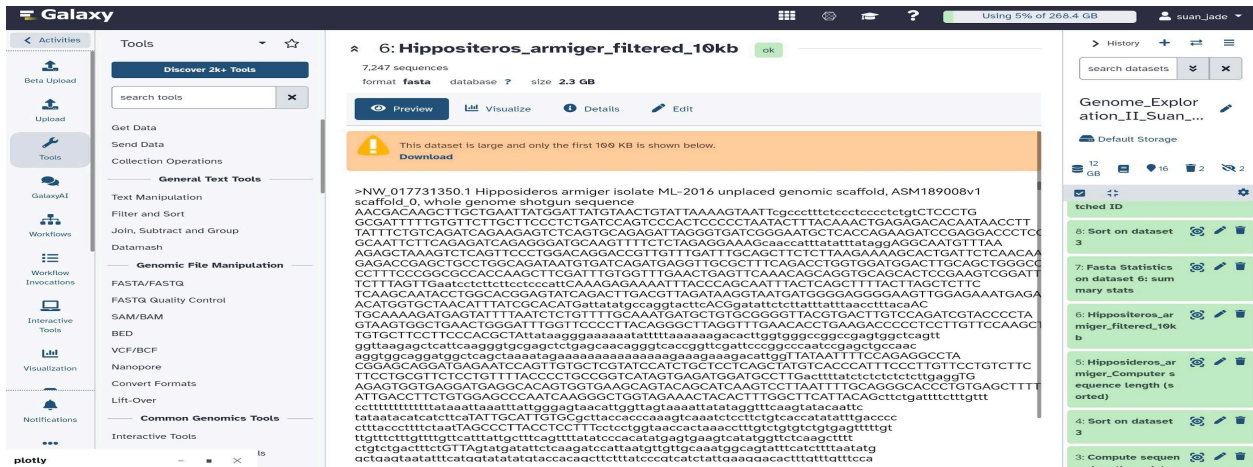


Figure 5. *Hippositeros armiger* Filtered 10kb Sequences Output using Filter Sequences by Length in Galaxy History

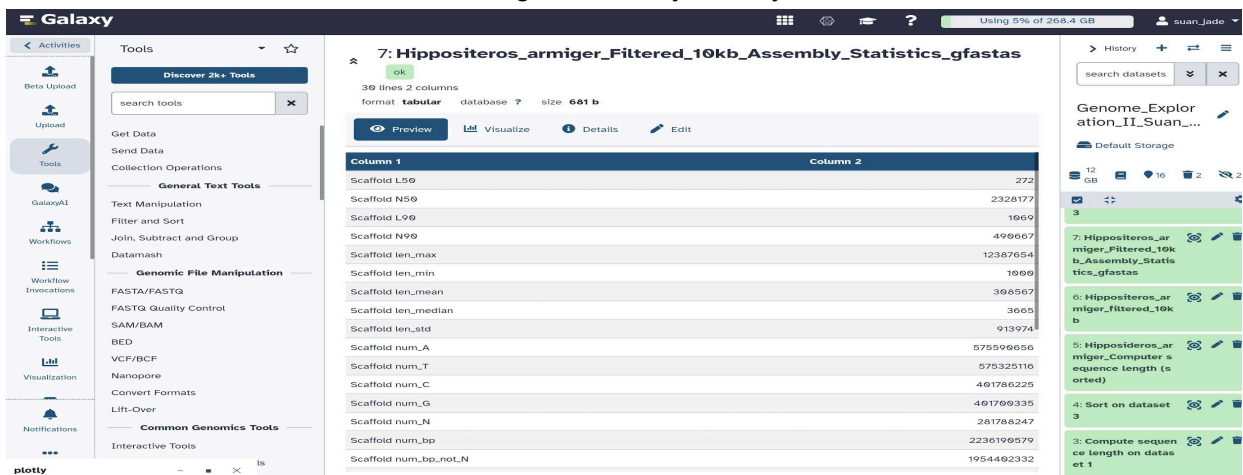


Figure 6. *Hippositeros armiger* Filtered 10kb Assembly Statistics Output rerun using gfastats in Galaxy History

- How many sequences were removed by the 10 kb filter?
 - 140 sequences were removed (7,387 - 7,247). This is a very small number, only about 1.9% of all scaffolds.
- Did the total assembly length decrease greatly or only slightly?
 - It decreased only slightly by just 135,735 bp out of over 2.2 billion bp, that's less than 0.01% of the total length. The short sequences that were removed are very small and barely contribute to the total genome size.
- Did N50 change? Why do you think it changed or stayed similar?
 - N50 stayed exactly the same at 2,328,177 bp. This happened because the sequences that were removed were all very short none of them were long enough to affect the N50 value.
- What does this tell you about the contribution of short sequences to your assembly?

- Short sequences contribute almost nothing to the total genome length even though there were 140 short scaffolds, they added up to barely any total sequence. Most of the genome is already in long, well-assembled scaffolds; this shows that the assembly is very high quality because very few short, fragmented pieces exist.

Part 5 - Small ORF Exploration

Metric	Value
Selected Sequence	NW_017732759.1
Sequence length	193,330 bp
Minimum ORF size used	300 bp
Number of ORF's found	173
Longest ORF	4,145 bp (NW_017732759.1_46)

[illegible]

Figure 7. *Hippositera armiger* Genome Original Uncompressed with Matched ID Output using the tool Filter sequence by ID in Galaxy History

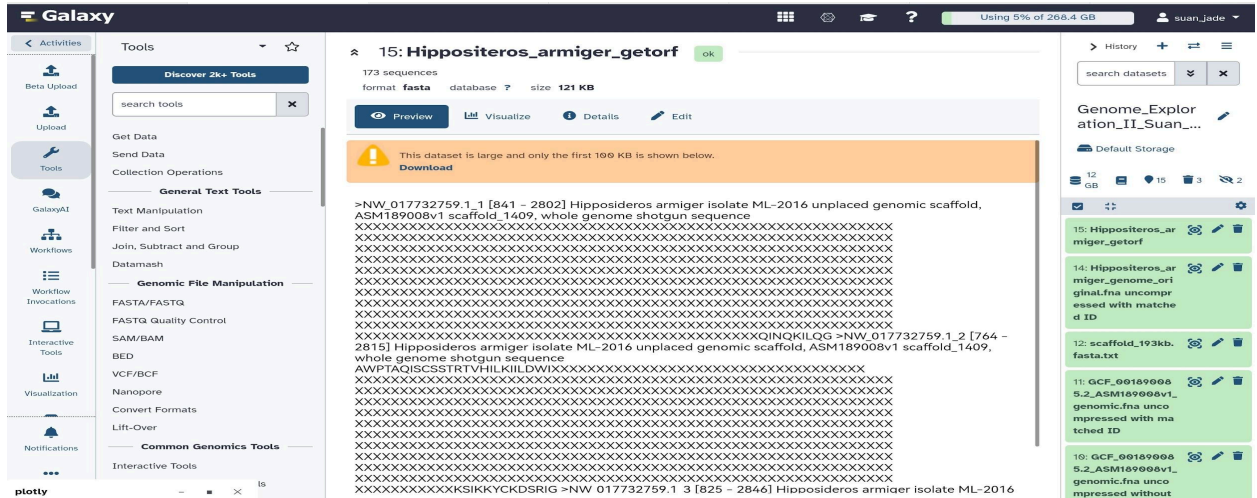


Figure 8. *Hipposetora armiger* ORF Prediction Output using getorf in Galaxy History

- How many ORFs were found in the selected sequence/region?
 - 173 open reading frames (ORFs) were detected in the 193,330 bp scaffold using a minimum ORF size of 300 bp.
- What was the longest ORF?
 - The longest ORF found was 4,145 bp in length, long enough to potentially encode a protein.
- Why should you NOT report every ORF as a real gene?
 - An ORF is only a stretch of DNA with no internal stop codon so it is not automatically a real gene. Many ORFs happen just by random chance. To confirm it is an actual gene, we need extra proof like matching RNA data, similarity to known genes or promoter sequences. We cannot treat every ORF as a gene without this extra evidence.

Part 6 - Short Biological Interpretation

The genome assembly of *Hipposideros armiger* is relatively contiguous and not highly fragmented. With an N50 value of 2,328,177 bp and a maximum scaffold length of 12,387,654 bp show that large sections of the genome are assembled into long, well-connected sequences, indicating good assembly quality. Filtering sequences shorter than 10 kb removed only 140 out of 7,387 scaffolds, and reduced total length by less than 0.01%, which means short sequences make almost no contribution to the total genome size. The GC content is 35.93%, a typical value for bat and other mammalian genomes. The ORF exploration found 173 candidate open reading frames, with the longest being 4,145 bp this means there are likely regions that could code for proteins. However, I know an ORF alone does not prove a real gene exists. It could just be a random sequence, so further evidence like matching RNA data or similarity to known genes would be needed to confirm these are actual functional genes.

Part 7 - Galaxy Reproducibility and GitHub Documentations

GALAXY	GITHUB
Student's Name: Suan, Jade Angela B.	Student's Name: Suan, Jade Angela B.
Email: jadeangela.suan@gmail.com	Email: jadeangela.suan@gmail.com
Username: suan_jade	Username: Jayeydd
Link: https://usegalaxy.org/u/suan_jade/h/genome-exploration-ii-suan-hipposideros-armiger	Link: https://github.com/Jayeydd/assignment_02_genome_exploration

Using 5% of 268.4 GBsuan_jade

History

search datasets

Genome_Exploration_II_Suan_Hipposideros_armiger

Default Storage

12 GB

992

15: Hippositeros_armiger_getorf

14: Hippositeros_armiger_genome_original.fna uncompressed with matched ID

7: Hippositeros_armiger_Filtered_10kb_Assembly_Statistics_gfastas

6: Hippositeros_armiger_filtered_10kb

5: Hipposideros_armiger_Computer sequence length (sorted)

3: Hippositeros_armiger_ Compute sequence length

2: Hippositeros_armiger_Assembly_Statistics_gfastats

1: Hipposideros_armiger_genome_original.fna.gz

Figure 9. Workflow History of *Hipposideros armiger* Genome Exploration in Galaxy

Jayeydd / assignment_02_genome_exploration

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JayeyddAdd files via upload642fd89 · 2 minutes ago43 Commits

Galaxy Link	Update Galaxy Link	1 hour ago
PART 2_ ASSEMBLY STATISTICS USING gfastats ...	Add files via upload	48 minutes ago
PART 3 - SEQUENCE LENGTH ANALYSIS (SORTE...	Add files via upload	33 minutes ago
PART 3 - SEQUENCE LENGTH ANALYSIS (TABUL...	Add files via upload	42 minutes ago
PART 3 - SEQUENCE LENGTH ANALYSIS (HISTO...	Add files via upload	29 minutes ago
PART 4 - SIMPLE LENGTH FILTERING EXPERIME...	Add files via upload	18 minutes ago
PART 4 - SIMPLE LENGTH-FILTERING EXPERIME...	Add files via upload	15 minutes ago
PART 5- SMALL ORF EXPLORATION (PREDICTIO...	Add files via upload	4 minutes ago
PART 5- SMALL ORF EXPLORATION.pdf	Add files via upload	11 minutes ago
PART1_GENOME UPLOAD.pdf	Add files via upload	52 minutes ago
README.md	Update README.md	1 hour ago
Results Table.pdf	Add files via upload	2 minutes ago

About

Genome exploration - Hippisideros armiger

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Jayeydd

Figure 10. GitHub Repository for the Activity